

Deepak Mallampati

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Kent, OH - US-based, F-1 OPT (open to sponsorship)

PROFESSIONAL SUMMARY

Applied AI engineer who ships production integration code, not just demos. Built and published a multi-provider browser extension (Manga Lens, Chrome Web Store) integrating four AI vision providers behind one common interface with provider-specific payload handling and seven-day translation caching - the same integration discipline E2B's enterprise customers need when connecting sandboxes to their AI agent stacks. Packaged LLM inference as Python/FastAPI services in Docker (~30% defect reduction post-deploy), shipped agentic LLM workflows with grounded reasoning and tool discipline (>30% hallucination reduction), and ran T-SQL/Jenkins CI/CD in a compliance-sensitive industrial environment at Emerson (~35-40% release cycle time gain). Master's in Computer Science (Kent State). Open to relocating to San Francisco. US-based, F-1 OPT.

CORE COMPETENCIES

Python • TypeScript • FastAPI • Docker • RESTful APIs & SDK Integration • Multi-Provider AI Integration • Customer-Embedded Delivery • CI/CD (Jenkins) • PostgreSQL / T-SQL • LLM Agents & Orchestration • RAG & Evaluation Harnesses

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present | USA

Jr. AI/ML Engineer Trainee - Production AI Integration, Healthcare

- Packaged ML/LLM inference as **FastAPI/Flask** RESTful services, containerized with **Docker**, with structured logging and load simulation - the same SDK + container pattern E2B customers use to integrate sandboxed runtimes; reduced post-deployment defects ~**30%**.
- Built **agentic LLM workflows** with structured reasoning, tool discipline, and grounding rules; improved agent response stability ~**25%** - directly relevant to AI agents that need secure sandboxed execution.
- Built **RAG systems** with recursive semantic chunking and transformer embeddings; improved contextual retrieval precision ~**35%** and reduced irrelevant retrieval >**30%**.
- Maintained **HIPAA-conscious data governance**: de-identification, data lineage, evaluation audit trails, stakeholder-facing system-limitation docs - directly transferable to SOC 2 / GDPR enterprise conversations.
- Shipped predictive ML pipelines (scikit-learn, XGBoost); improved recall **15-20%** on high-risk categories while holding precision >**90%**.

Suvidha Foundation

Jan 2022 - Mar 2022 | India

Machine Learning Engineer - Applied NLP, Video Analytics

- Built a transformer-based summarization system over 5,000+ sessions; hierarchical long-context summarization cut manual review time **60-70%**.
- Implemented **document Q&A with RAG** (semantic chunking + embedding retrieval); reduced hallucinated outputs ~**30%**, raised answer accuracy >**85%**.
- Deployed the stack as a **Flask** API with a lightweight web interface for non-technical users.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023 | India

Database & DevOps Performance Engineer (Intern) - Industrial / Oil & Gas Enterprise ERP

- Built **CI/CD pipelines (Jenkins)** for version-controlled SQL and stored-procedure deploys with structured release validation and rollback checkpoints; reduced deployment errors >**30%**, improved release cycle time ~**35-40%** - the same release discipline customer BYOC deployments need.
- Built **SQL Server DMV + Grafana** performance dashboards to track long-running queries, deadlocks, CPU/I/O contention; cut incident recurrence ~**25%**, improved RCA speed ~**30%**.
- Supported **RBAC + audit logging** for financial modules in a compliance-sensitive oil & gas environment - adjacent to enterprise SSO/IAM work.
- Tuned T-SQL stored procedures; cut query execution time ~**20%** and data retrieval latency ~**25%**.

PROJECTS

Manga Lens - Chrome Extension Integrating 4 AI Vision Providers (shipped, Chrome Web Store)

Full-stack browser extension built independently - Manifest V3, content scripts, service workers, **captureVisibleTab** for panel capture. Integrates **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v local) behind one interface, with **provider-specific payload handling** (WebP for cloud, JPEG for local Ollama to avoid CUDA crash), seven-day translation cache, per-domain selector configs for 29 sites, privacy policy, narrowed host permissions.

TypeScript, Manifest V3, service workers, multi-provider AI integration, payload abstraction

Agentic Healthcare Claims Pipeline - Schema-Contracted Multi-Agent System

Multi-agent pipeline (Intake → Validation → Consistency → Duplicate Detection → Fraud Risk) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy docs, ANN duplicate detection, explainable risk scoring with **audit-ready reasoning traces** - the same reliability discipline enterprise AI agent deployments need.

Multi-agent, JSON schema contracts, RAG, vector search, audit traces

Dream Decoder - Multimodal Creative Intelligence Platform

Full-stack **FastAPI + React/TypeScript/Vite/Tailwind** app coordinating multimodal APIs (Perplexity Sonar + Replicate image models). Intermediate **structured prompt transformation layers** improved contextual alignment **~30%** and first-pass image success **~25-30%**.

FastAPI, TypeScript, Vite, structured prompt orchestration

Driver Drowsiness Detection - Real-Time YOLOv8 with Adaptive Throttling

Replaced a two-stage CNN pipeline with a **unified YOLOv8** model; reduced inference latency **~30%**. Sliding-window confidence aggregation suppressed blink-driven false positives (**~25% reduction**), adaptive frame skipping for production stability under load.

YOLOv8, OpenCV, real-time CV, latency engineering

EDUCATION

Master's Degree, Computer Science - Kent State University, Ohio, USA

Aug 2023 - May 2025

SKILLS

Languages & Frameworks: Python, TypeScript, FastAPI, Flask, SQL (PostgreSQL, T-SQL, SQLite), React, Node.js

Infra & DevOps: Docker, Jenkins, CI/CD, RESTful APIs & SDK integration, observability/logging, Grafana, SQL Server DMVs

LLM & GenAI: Production LLM serving, RAG, agentic workflows, structured outputs, prompt engineering, evaluation pipelines, grounding, embeddings, vector search

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, LangChain, LlamaIndex

Enterprise / Compliance: HIPAA-conscious governance, RBAC, audit logging, release validation, rollback checkpoints